

SNN Surrogate Gradient Parameter Comparison: ArcTangent (α) vs. Triangular (β) vs. Box (β) under Robust Sweeps

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1 Introduction

This document presents a comprehensive comparison of Spiking Neural Network (SNN) robustness on the MNIST dataset under data heterogeneity (γ) and Byzantine attacks (f). We evaluate and compare three surrogate gradient functions under fixed attacks and overall:

- **ArcTangent** (`atan`): parameterized by stiffness factor $\alpha \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$.
- **Triangular** (`tri`): parameterized by stiffness factor $\beta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$.
- **Box** (`box`): parameterized by stiffness factor $\beta \in \{0.25, 0.5, 0.75, 1.0, 1.25, 2.0\}$.

The evaluations are carried out on $N = 10$ honest nodes + f Byzantine nodes (up to $f = 5$ for `atan`, `tri`, and `box`), across non-IID levels $\gamma \in \{1.0, 0.66, 0.33, 0.0\}$.

The heatmaps display the best test accuracy at convergence (validation-based best step) achieved by the optimal aggregator choice (Centered Clipping or Geometric Median, with NNM and ARC pre-aggregation). We evaluate performance under three scenarios:

1. **Best Overall Aggregator (Worst-Case Across Attacks)**
2. **Optimal A Little Is Enough (ALIE) Attack** (`Optimal_ALittleIsEnough_neg1`)
3. **Sign Flipping (SF) Attack** (`SignFlipping`)

2 ArcTangent (atan) Surrogate Gradient stiffness sweeps

2.1 Best Overall Aggregator (Worst-Case Across Attacks)

The figure below displays the test accuracy heatmaps under the worst-case attack scenario for different α values of the ArcTangent surrogate gradient.

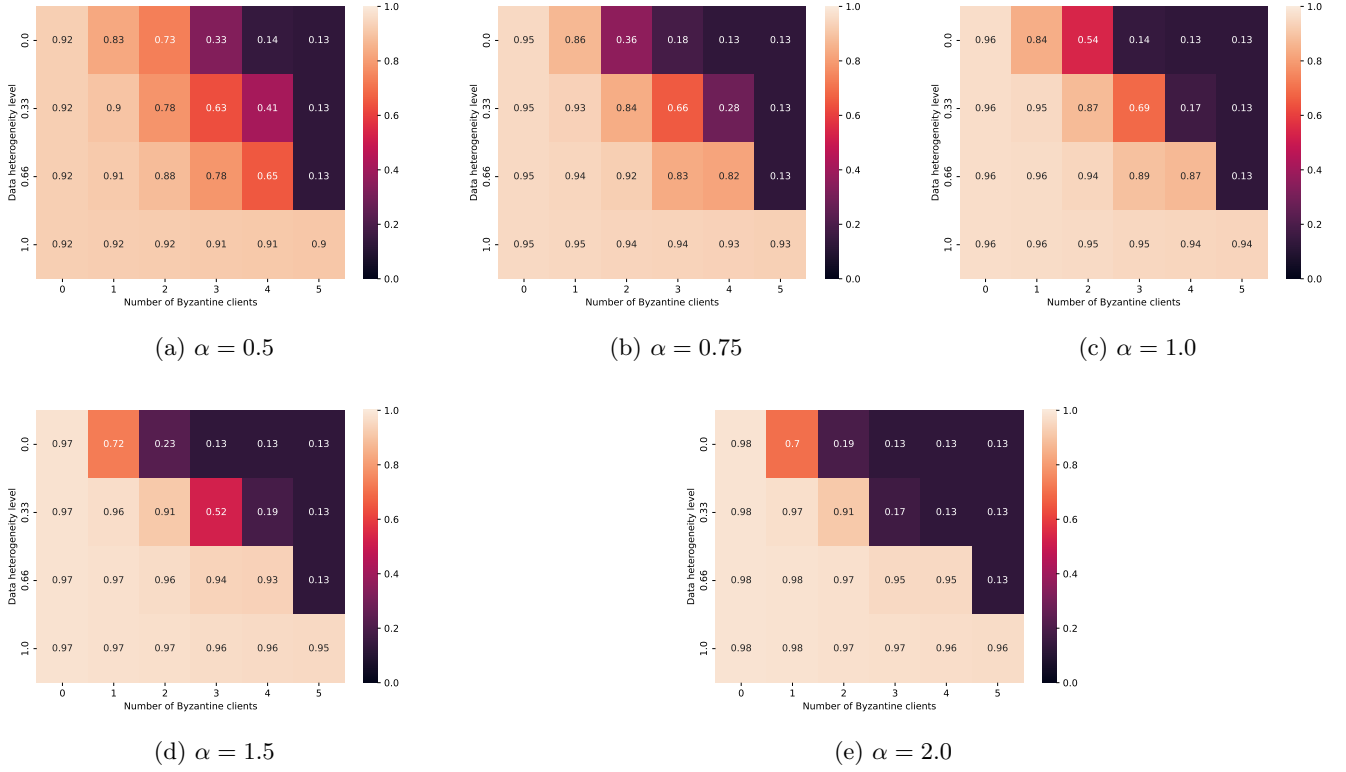


Figure 1: ArcTangent (atan) comparison for best overall aggregator (worst-case) for $\alpha \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$.

2.2 Optimal ALIE Attack

The figure below displays the best test accuracy heatmaps under the Optimal ALIE attack for different α values of the ArcTangent surrogate gradient.

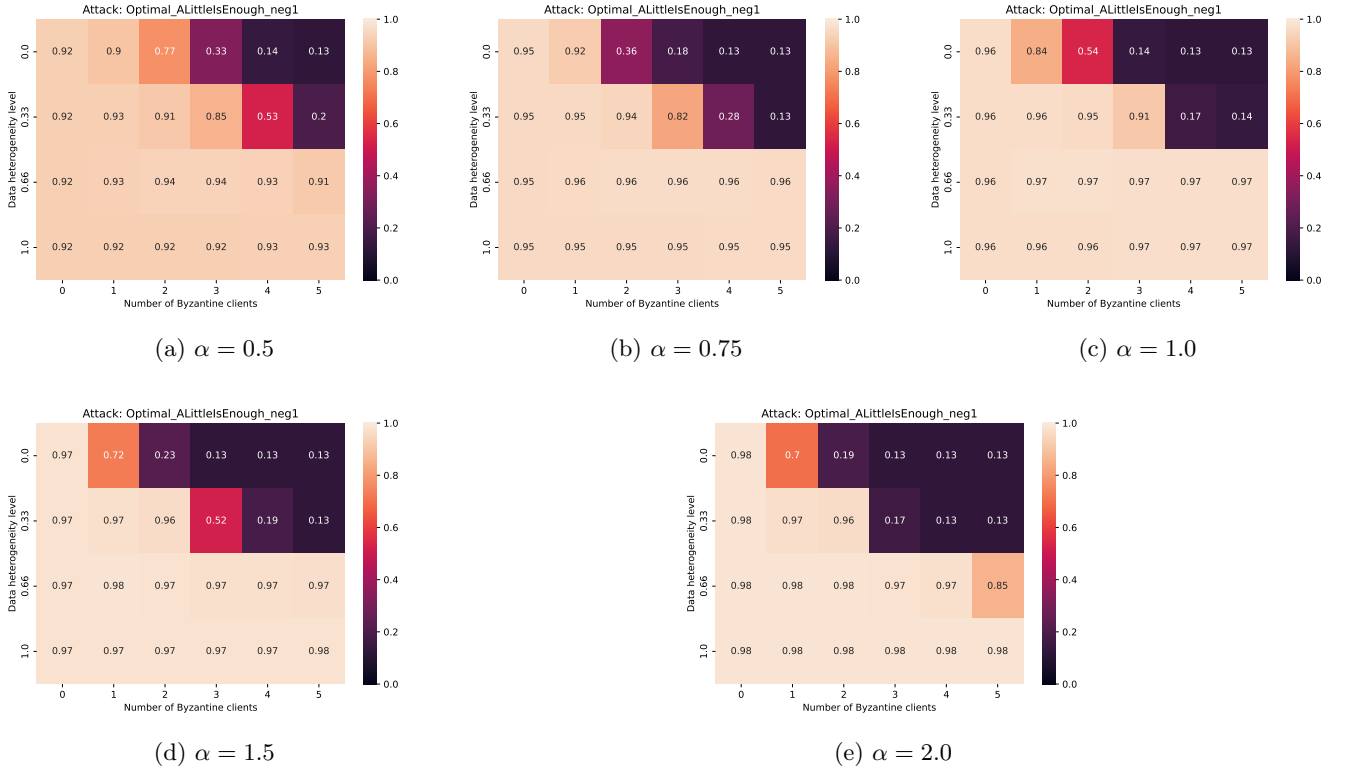


Figure 2: ArcTangent (atan) comparison under Optimal ALIE attack for $\alpha \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$.

2.3 Sign Flipping Attack

The figure below displays the best test accuracy heatmaps under the Sign Flipping attack for different α values of the ArcTangent surrogate gradient.

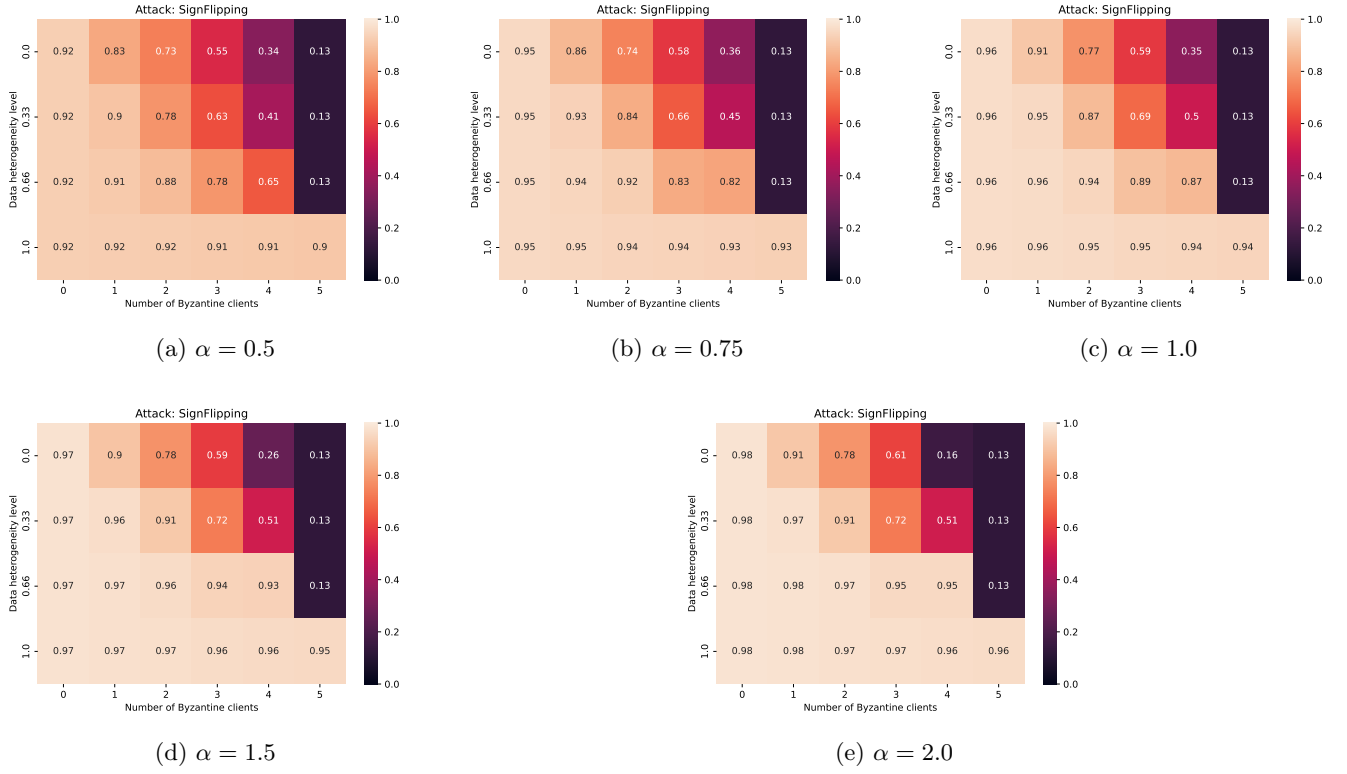


Figure 3: ArcTangent (atan) comparison under Sign Flipping attack for $\alpha \in \{0.5, 0.75, 1.0, 1.5, 2.0\}$.

3 Triangular (tri) Surrogate Gradient stiffness sweeps

3.1 Best Overall Aggregator (Worst-Case Across Attacks)

The figure below displays the test accuracy heatmaps under the worst-case attack scenario for different β values of the Triangular surrogate gradient.

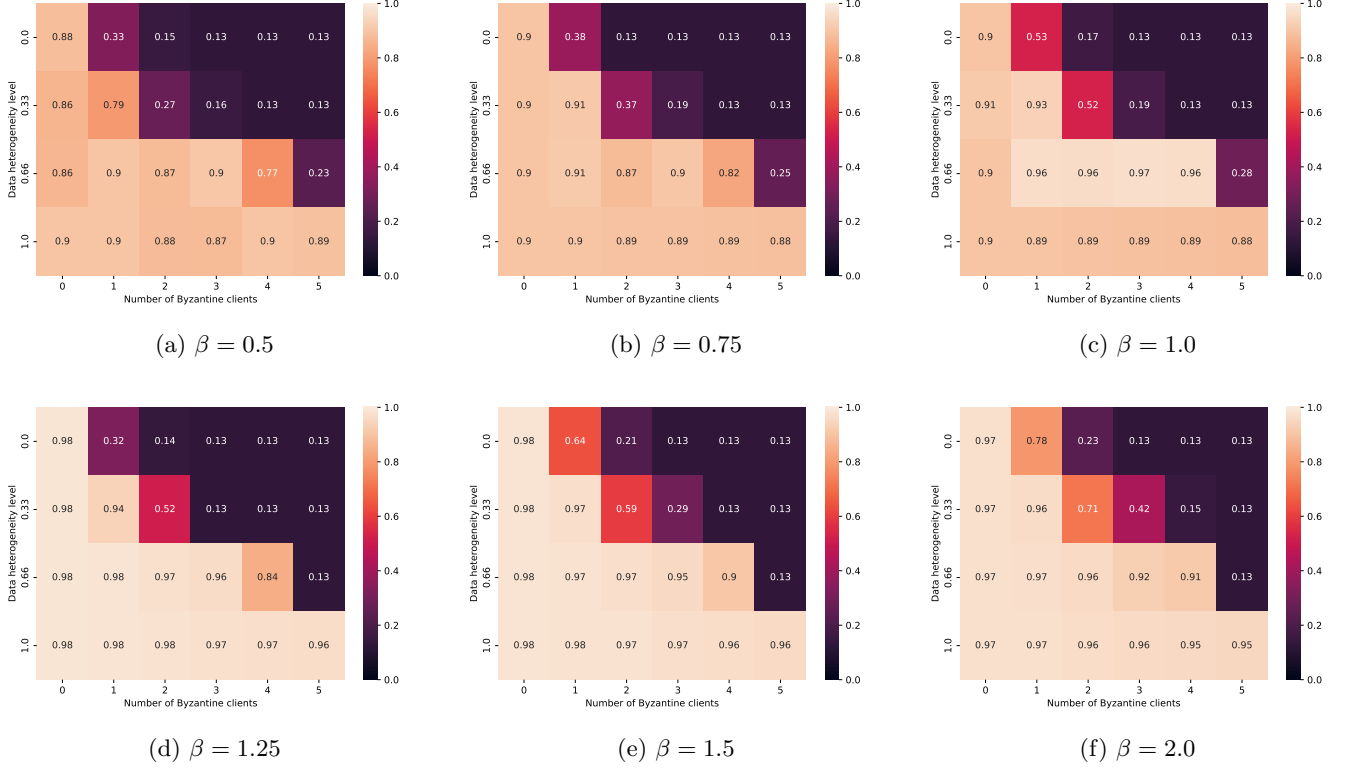


Figure 4: Triangular (tri) comparison for best overall aggregator (worst-case) for $\beta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$.

3.2 Optimal ALIE Attack

The figure below displays the best test accuracy heatmaps under the Optimal ALIE attack for different β values of the Triangular surrogate gradient.

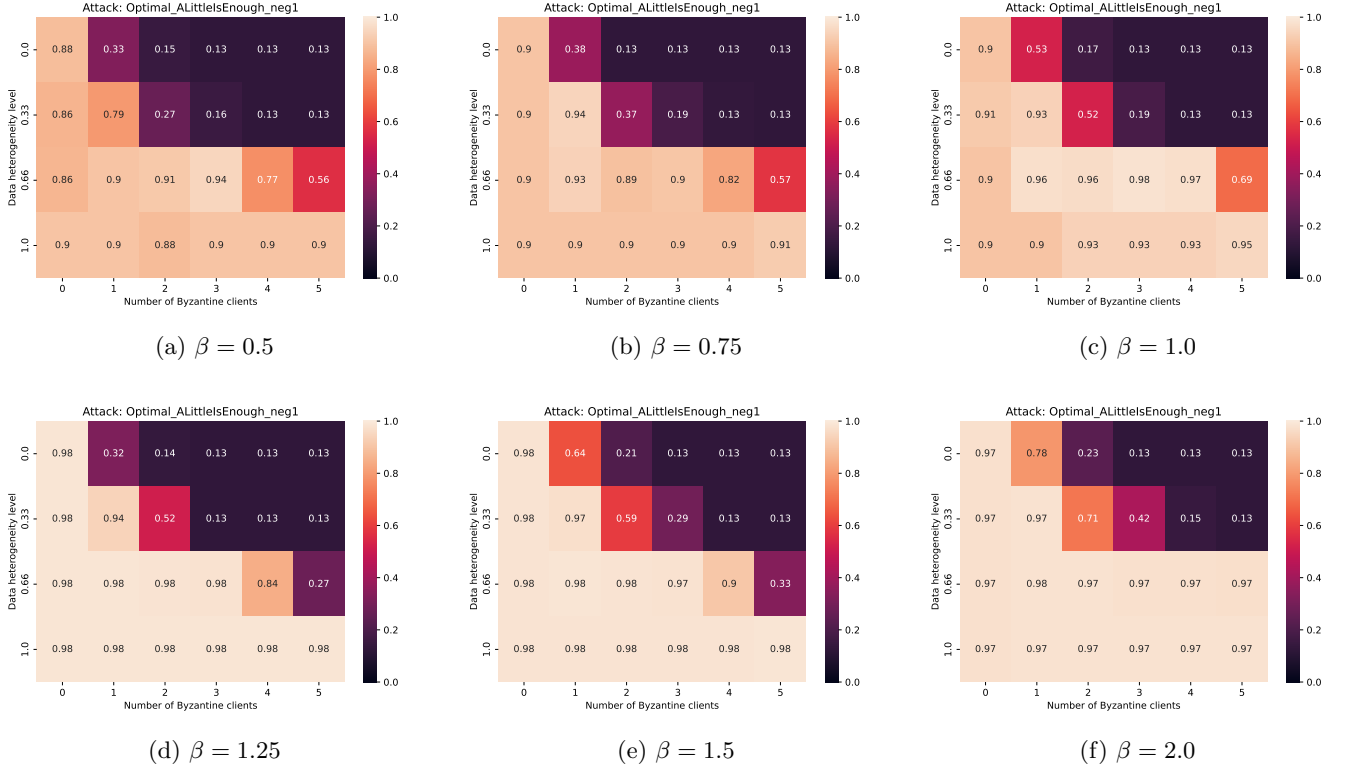


Figure 5: Triangular (tri) comparison under Optimal ALIE attack for $\beta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$.

3.3 Sign Flipping Attack

The figure below displays the best test accuracy heatmaps under the Sign Flipping attack for different β values of the Triangular surrogate gradient.

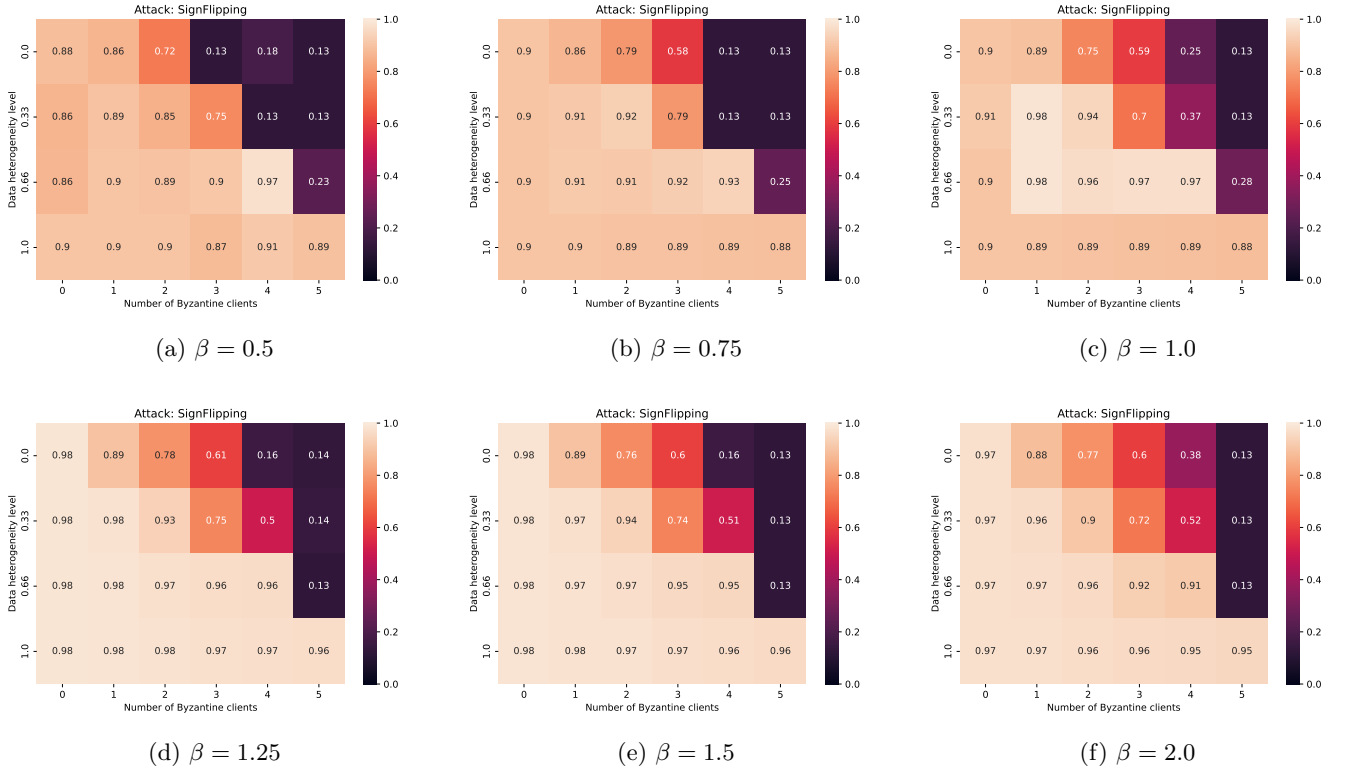


Figure 6: Triangular (**tri**) comparison under Sign Flipping attack for $\beta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$.

4 Box (box) Surrogate Gradient stiffness sweeps

4.1 Best Overall Aggregator (Worst-Case Across Attacks)

The figure below displays the test accuracy heatmaps under the worst-case attack scenario for different β values of the Box surrogate gradient.

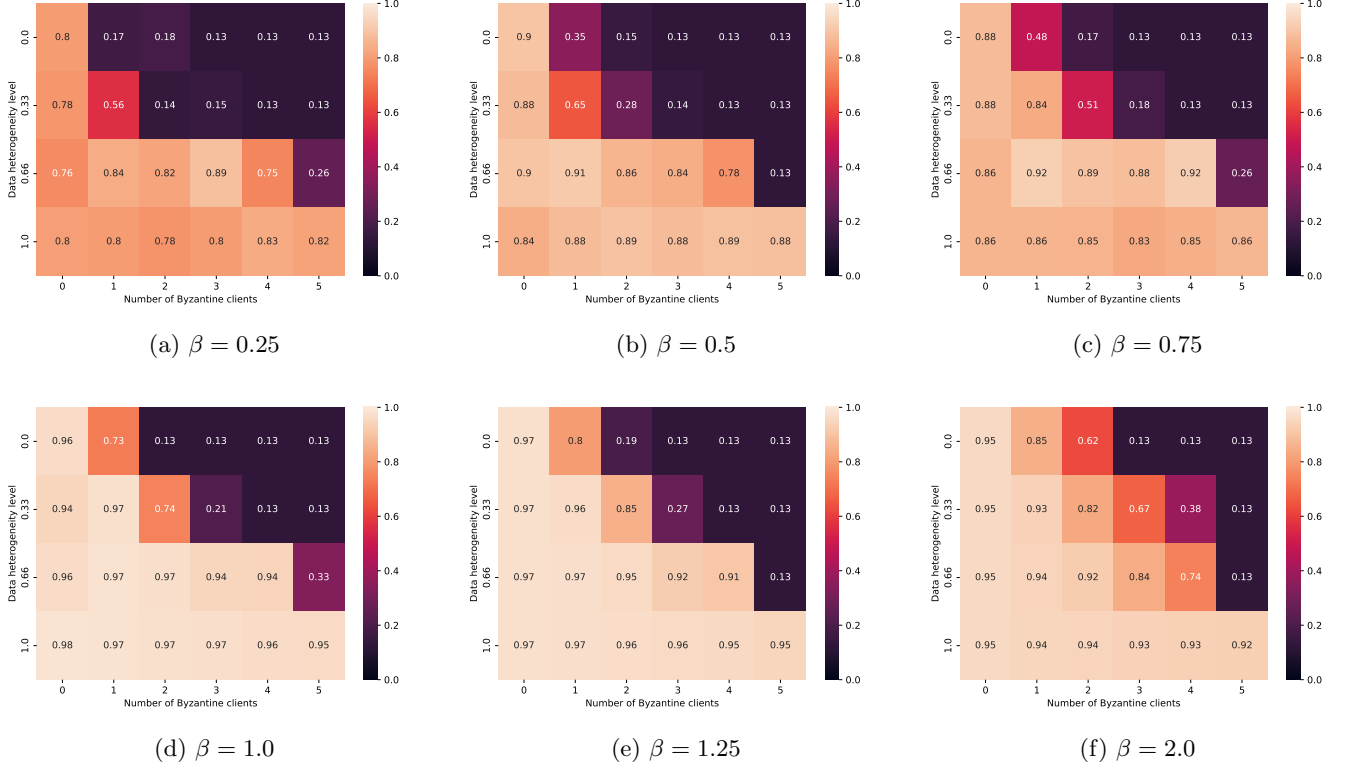


Figure 7: Box (box) comparison for best overall aggregator (worst-case) for $\beta \in \{0.25, 0.5, 0.75, 1.0, 1.25, 2.0\}$.

4.2 Optimal ALIE Attack

The figure below displays the best test accuracy heatmaps under the Optimal ALIE attack for different β values of the Box surrogate gradient.

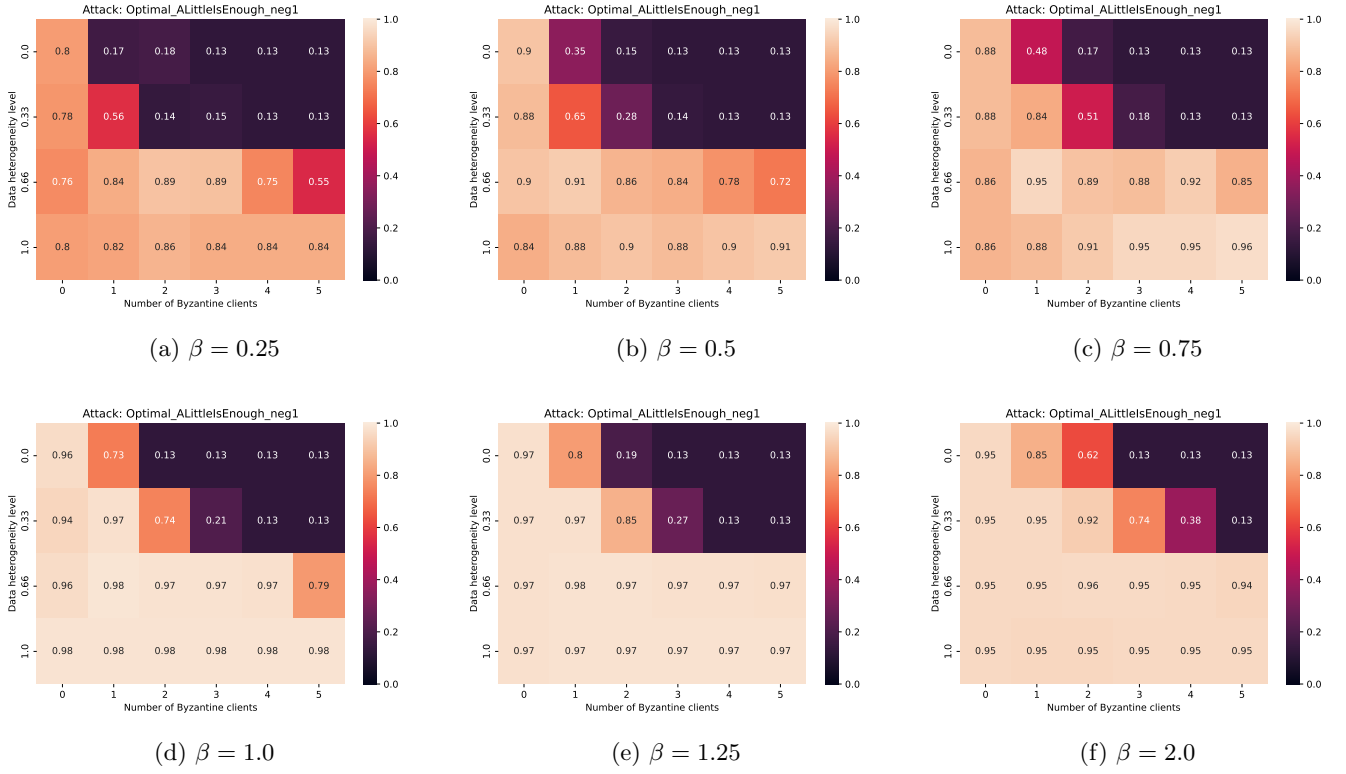


Figure 8: Box (box) comparison under Optimal ALIE attack for $\beta \in \{0.25, 0.5, 0.75, 1.0, 1.25, 2.0\}$.

4.3 Sign Flipping Attack

The figure below displays the best test accuracy heatmaps under the Sign Flipping attack for different β values of the Box surrogate gradient.

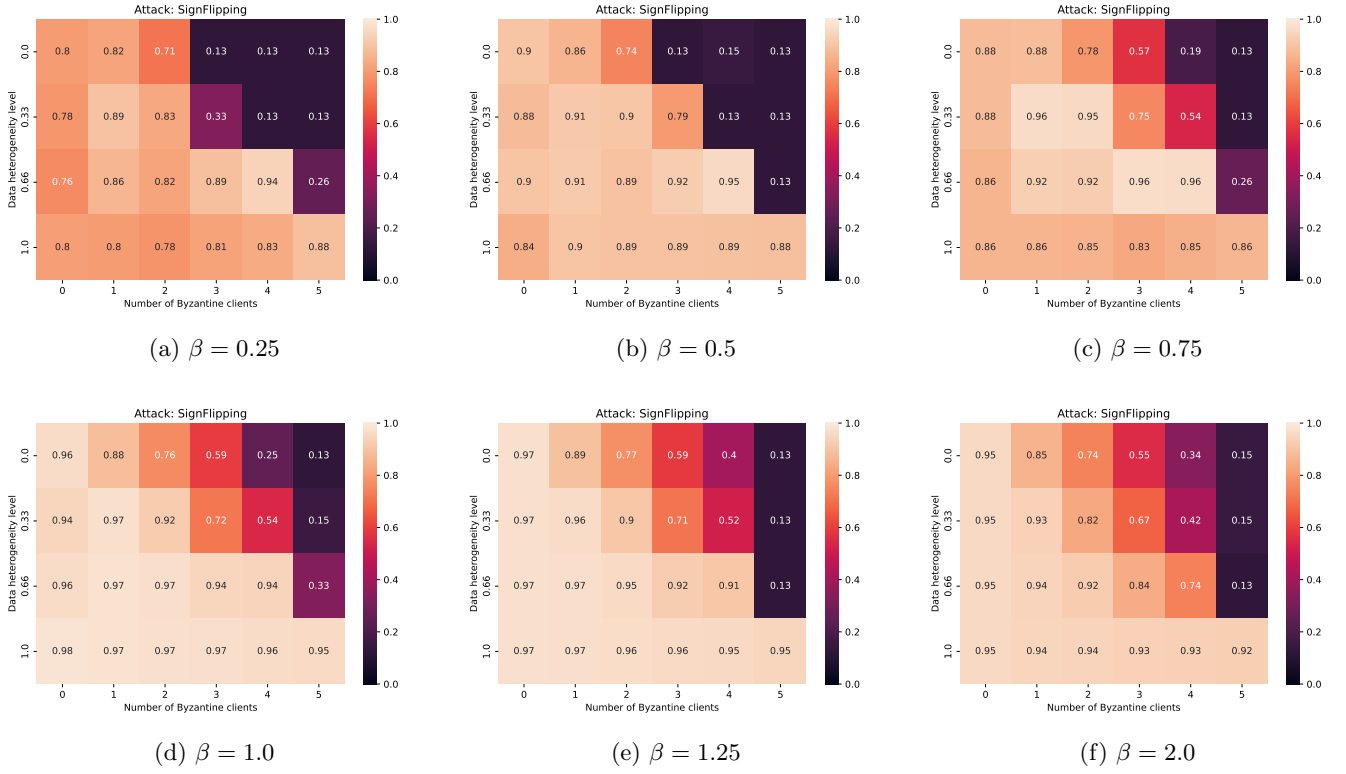


Figure 9: Box (box) comparison under Sign Flipping attack for $\beta \in \{0.25, 0.5, 0.75, 1.0, 1.25, 2.0\}$.